

Complete EDA Quiz with Answers and Explanations

1. Why is outlier detection important in EDA?

- A) To upload data to a database
- B) To identify and manage anomalous data points that may skew analysis
- C) To create visualizations
- D) To fill missing values

Correct Answer: D) To fill missing values

Explanation: Concept related to option D.

2. How can you handle imbalanced data in a dataset?

- A) Data scaling
- B) Log transformation
- C) Resampling techniques
- D) Data visualization

Correct Answer: A) Data scaling

Explanation: Concept related to option A.

3. Which plot is used to visualize the relationships between more than two variables?

- A) Pie Chart
- B) Scatter Plot
- C) Pair Plot
- D) Bar Chart

Correct Answer: D) Bar Chart

Explanation: Concept related to option D.

4. What does the p-value signify in hypothesis testing?

- A) The probability of observing the data given that the null hypothesis is true
- B) The probability of the null hypothesis being true
- C) The probability of the alternative hypothesis being true
- D) The probability of observing the data given that the alternative hypothesis is true

Correct Answer: Not marked

Explanation: Not available.

5. Which test is commonly used to compare the means of two groups?

- A) Chi-square test
- B) ANOVA
- C) Correlation test (X
- D) t-test

Correct Answer: Not marked

Explanation: Not available.

6. Which method is commonly used to test the normality of a dataset?

- A) Chi-square test
- B) Shapiro-Wilk Test
- C) Correlation test
- D) t-test

Correct Answer: D) t-test

Explanation: Concept related to option D.

7. How can you describe the skewness of a dataset?

- A) The measure of asymmetry of the distribution
- B) The measure of central tendency
- C) The measure of peakness of the distribution
- D) The measure of spread of the distribution

Correct Answer: D) The measure of spread of the distribution

Explanation: Concept related to option D.

8. Which method is used to calculate the correlation between variables?

- A) rel() method
- B) correlation() method
- C) relation() method
- D) corr() method

Correct Answer: B) correlation() method

Explanation: Concept related to option B.

9. What is the alternative hypothesis in hypothesis testing?

- A) A statement of effect or difference

- B) A statement of causation
- C) A statement of no effect or no difference
- D) A statement of correlation

Correct Answer: D) A statement of correlation

Explanation: Concept related to option D.

10. Which measure provides the spread of data around the mean?

- A) Mode
- B) Median
- C) Standard Deviation
- D) Mean ()

Correct Answer: Not marked

Explanation: Not available.

11. What is the main objective of normalization in data processing?

- A) To scale the data to a specific range
- B) To calculate the mean
- C) To create visualizations
- D) To fill missing values

Correct Answer: D) To fill missing values

Explanation: Concept related to option D.

12. What is the purpose of one-hot encoding in handling categorical data?

- A) To convert categorical data into numerical format
- B) To reduce the size of the dataset X
- C) To create visualizations
- D) To upload data to a database

Correct Answer: Not marked

Explanation: Not available.

13. Which method is commonly used for non-linear dimensionality reduction?

- A) t-Distributed Stochastic Neighbor Embedding (t-SNE)
- B) Principal Component Analysis (PCA)
- C) K-means clustering

D) Linear Discriminant Analysis (LDA)

Correct Answer: D) Linear Discriminant Analysis (LDA)

Explanation: Concept related to option D.

14. Which method is commonly used to calculate the covariance between variables?

A) var() method

B) cov() method

C) std() method

D) cor() method

Correct Answer: D) cor() method

Explanation: Concept related to option D.

15. What is the significance of the variance in a dataset?

A) It provides a measure of the spread of data

B) It provides a measure of correlation

C) It provides a measure of skewness

D) It provides a measure of central tendency

Correct Answer: Not marked

Explanation: Not available.

16. Which clustering method does not require the number of clusters to be specified a priori?

A) Linear Discriminant Analysis (LDA)

B) K-means clustering (X)

C) Hierarchical clustering

D) DBSCAN

A) t-SNE

B) Principal Component Analysis (PCA)

C) Linear Discriminant Analysis (LDA)

Correct Answer: C) Hierarchical clustering

Explanation: Concept related to option C.

17. Which method is commonly used to identify the optimal number of clusters in a dataset?

D) Elbow Method

Correct Answer: Not marked

Explanation: Not available.

18. Which plot is used to visualize the relationship between a numerical and a categorical variable?

- A) Scatter Plot
- B) Box Plot
- C) Pie Chart
- D) Histogram

Correct Answer: D) Histogram

Explanation: Concept related to option D.

19. How can you determine the strength of a relationship between two variables?

- A) By calculating the median
- B) By calculating the mean
- C) By calculating the correlation coefficient
- D) By creating visualizations

Correct Answer: B) By calculating the mean

Explanation: Concept related to option B.

20. What is the benefit of using a violin plot in EDA?

- A) To fill missing values
- B) To create databases
- C) To upload data to a database
- D) To visualize the distribution and density of data

Correct Answer: Not marked

Explanation: Not available.

21. What is the significance of the median in a dataset?

- A) It represents the spread of data

Correct Answer: Not marked

Explanation: Not available.

22. What does the term 'Bivariate Analysis' refer to in EDA?

- A) Analysis of more than two variables
- B) Analysis of two variables to determine relationships

C) Analysis of the entire dataset

D) Analysis of a single variable

Correct Answer: Not marked

Explanation: Not available.

23. Why might you use logistic regression in EDA?

A) To create visualizations

B) To upload data to a database

C) To model binary outcomes

D) To fill missing values

Correct Answer: Not marked

Explanation: Not available.

24. What is the main challenge in handling categorical data in EDA?

A) Uploading data to a database

B) Creating visualizations

C) Encoding it in a way suitable for machine learning models

D) Reducing the size of the dataset

Correct Answer: Not marked

Explanation: Not available.

25. Why might you perform a log transformation on a dataset?

A) To fill missing values

B) To create visualizations

C) To upload data to a database

D) To manage skewness and spread in the data

Correct Answer: Not marked

Explanation: Not available.

26. Which plot is commonly used to visualize hierarchical clustering?

A) Scatter plot

B) Pie chart

C) Dendrogram

D) Bar chart

Correct Answer: Not marked

Explanation: Not available.

27. Which method is commonly used to impute missing values in a dataset?

- A) Mean imputation
- B) Log transformation
- C) Data scaling
- D) Data visualization

Correct Answer: Not marked

Explanation: Not available.

28. Which plot is commonly used to visualize the distribution of a continuous variable?

- A) Pie Chart
- B) Scatter Plot
- C) Bar Chart
- D) Histogram

Correct Answer: Not marked

Explanation: Not available.

29. What does the term 'Kurtosis' signify in a dataset?

- A) The measure of the central tendency

Correct Answer: A) The measure of the central tendency

Explanation: Concept related to option A.

30. What is the primary goal of correlation analysis?

- A) To categorize data
- B) To identify relationships between variables
- C) To identify missing values
- D) To predict future data points

Correct Answer: Not marked

Explanation: Not available.

31. What does the term 'Feature Extraction' refer to in EDA?

- A) Uploading data to a database

B) Filling missing values

C) Creating new features from existing features to better capture the underlying data patterns

D) Creating visualizations

Correct Answer: Not marked

Explanation: Not available.

32. What is the purpose of data transformation in EDA?

A) To achieve a better fit for modeling

B) To create visualizations

C) To reduce the size of the dataset

D) To upload data to a database

Correct Answer: Not marked

Explanation: Not available.

33. What is the purpose of dimensionality reduction in EDA?

A) To create visualizations

B) To reduce the number of variables and simplify analysis

C) To fill missing values

D) To upload data to a database

Correct Answer: Not marked

Explanation: Not available.

34. What does the term 'Encoding' refer to in handling categorical data?

A) Reducing the size of the dataset

B) Converting categorical data into a numerical format

C) Creating visualizations

D) Uploading data to a database

A) To create visualizations

Correct Answer: Not marked

Explanation: Not available.

35. Why is feature engineering important in EDA?

B) To create meaningful variables for modeling

C) To upload data to a database

D) To reduce the size of the dataset

Correct Answer: Not marked

Explanation: Not available.

36. What is the benefit of scaling data in EDA?

A) To upload data to a database

B) To fill missing values

C) To ensure that variables are compared on a similar scale

D) To create visualizations

Correct Answer: Not marked

Explanation: Not available.

37. Which method is commonly used to handle missing categorical data?

A) Standard Deviation imputation

B) Mean imputation

C) Median imputation

D) Mode imputation

Correct Answer: Not marked

Explanation: Not available.

38. Why might you use a box plot in EDA?

A) To upload data to a database

B) To visualize the distribution and identify outliers

C) To create databases

D) To fill missing values

Correct Answer: Not marked

Explanation: Not available.

39. What is the primary objective of Exploratory Data Analysis (EDA)?

A) To understand the underlying patterns in the data

B) To implement machine learning algorithms

C) To create databases

D) To build predictive models

Correct Answer: Not marked

Explanation: Not available.

40. What is the primary goal of the Chi-square test in EDA?

- A) To upload data to a database
- B) To create visualizations
- C) To test the association between categorical variables
- D) To fill missing values

Correct Answer: Not marked

Explanation: Not available.

41. What is the benefit of encoding categorical data into numerical data?

- A) Creating visualizations
- B) Uploading data to a database
- C) To prepare the data for machine learning algorithms
- D) To fill missing values

Correct Answer: D) To fill missing values

Explanation: Concept related to option D.

42. What is the purpose of a Pair Plot in EDA?

- A) To visualize the distribution of a single variable
- B) To visualize the distribution of single variables and relationships between two variables
- C) To visualize the spread of data
- D) To visualize the summary statistics of a dataset

Correct Answer: Not marked

Explanation: Not available.

43. Which method is commonly used to handle missing data?

- A) Imputation
- B) Normalization
- C) Correlation
- D) Standardization

Correct Answer: Not marked

Explanation: Not available.

44. What is a Quantile in a dataset?

- A) The average value
- B) The most common value
- C) Value below which a given percentage of data falls
- D) The spread of data

Correct Answer: Not marked

Explanation: Not available.

45. What does the term 'Outlier' signify in a dataset?

- A) The last data point in a dataset
- B) A data point significantly different from others
- C) The most common data point
- D) A data point in the middle of the range

Correct Answer: Not marked

Explanation: Not available.

46. What is the null hypothesis in hypothesis testing?

- A) A statement of no effect or no difference
- B) A statement of correlation
- C) A statement of causation
- D) A statement of effect or difference

Correct Answer: Not marked

Explanation: Not available.

47. Which method is used to calculate the correlation matrix?

- A) `corr()` method
- B) `correlation()` method
- C) `cov()` method
- D) `relation()` method

Correct Answer: Not marked

Explanation: Not available.

48. Which method is used to replace missing values in a dataset?

- A) `missing()` method

B) fillna() method

C) null() method

D) replace() method

Correct Answer: Not marked

Explanation: Not available.

49. Which measure is used to understand the spread of a dataset?

A) Variance

B) Mode

C) Median

D) Mean

Correct Answer: Not marked

Explanation: Not available.

50. In the Titanic dataset how would you find the number of passengers who survived?

A) titanic.sum('Survived')

B) titanic['Survived'].count()

C) titanic.count('Survived')

D) titanic['Survived'].sum()

Correct Answer: Not marked

Explanation: Not available.

51. Which plot is used to visualize the distribution of a categorical variable?

A) Line Plot

B) Bar Plot

C) Histogram

D) Scatter Plot

Correct Answer: Not marked

Explanation: Not available.

52. What is the purpose of binning or bucketing data in EDA?

A) To remove outliers

B) To group data into intervals

C) To find the median of the data

D) To find the mean of the data

Correct Answer: D) To find the mean of the data

Explanation: Concept related to option D.

53. What is the primary goal of forecasting models in time series analysis?

A) To predict future values based on past and present data

B) To upload data to a database

C) To fill missing values

D) To create visualizations

Correct Answer: B) To upload data to a database

Explanation: Concept related to option B.

54. What is the primary goal of statistical inference in EDA?

A) To upload data to a database

B) To fill missing values

C) To create visualizations

D) To make predictions or inferences about a population based on a sample

Correct Answer: Not marked

Explanation: Not available.

55. What does multivariate analysis allow you to do?

A) Calculate the mean

B) Analyze a single variable

C) Create visualizations

D) Analyze more than two variables to understand relationships

Correct Answer: Not marked

Explanation: Not available.

56. Why is time series analysis important in EDA?

A) To analyze data points collected or recorded at specific time intervals

B) To upload data to a database

C) To create visualizations

D) To reduce the size of the dataset

Correct Answer: Not marked

Explanation: Not available.

57. What is the primary purpose of advanced visualization techniques in EDA?

- A) To fill missing values
- B) To understand complex relationships in the data more clearly
- C) To create databases
- D) To upload data to a database

Correct Answer: Not marked

Explanation: Not available.

58. How are missing values represented in a Pandas DataFrame?

- A) Null
- B) O
- C) Missing
- D) NaN

Correct Answer: Not marked

Explanation: Not available.

59. What is the primary goal of forecasting in time series analysis?

- A) To upload data to a database
- B) To create visualizations
- C) To predict future values based on past observations
- D) To fill missing values

Correct Answer: A) To upload data to a database

Explanation: Concept related to option A.

60. What is the purpose of hypothesis testing in EDA?

- A) To create visualizations
- B) To make inferences about populations based on samples
- C) To upload data to a database
- D) To fill missing values

Correct Answer: Not marked

Explanation: Not available.

61. Which plot can be used to visualize the distribution of categorical data?

- A) Scatter Plot
- B) Histogram
- C) Bar Chart
- D) Box Plot

Correct Answer: Not marked

Explanation: Not available.

62. What does a Box Plot represent?

- A) The correlation between variables
- B) The relationship between two variables
- C) The summary statistics of a dataset including outliers
- D) The distribution of categorical data

Correct Answer: Not marked

Explanation: Not available.

63. What does the term 'Multivariate Analysis' refer to in EDA?

- A) Analysis of two variables
- B) Analysis of a single variable
- C) Analysis of summary statistics
- D) Analysis involving more than two variables

Correct Answer: Not marked

Explanation: Not available.

64. What is the primary goal of Linear Discriminant Analysis (LDA) in EDA?

- A) To fill missing values
- B) To find the linear combinations of features that best separate different classes
- C) To create visualizations
- D) To upload data to a database

Correct Answer: Not marked

Explanation: Not available.

65. What is the purpose of a scatter matrix plot in EDA?

- A) To upload data to a database

- B) To visualize pairwise relationships in a dataset
- C) To create single variable distributions
- D) To fill missing values

Correct Answer: A) To upload data to a database

Explanation: Concept related to option A.

66. Which plot is used to visualize the distribution of a dataset?

- A) Pie chart
- B) Bar chart
- C) Scatter plot
- D) Histogram

Correct Answer: Not marked

Explanation: Not available.

67. Which graph is used to display the relationship between two categorical variables?

- A) Histogram
- B) Stacked Bar Chart
- C) Scatter Plot
- D) Box Plot

Correct Answer: Not marked

Explanation: Not available.

68. Why are heatmaps used in EDA?

- A) To fill missing values
- B) To create databases
- C) To upload data to a database
- D) To visualize the correlation or other relationships between variables

Correct Answer: Not marked

Explanation: Not available.

69. Which measure is commonly used to understand the dispersion or spread in a dataset?

- A) Median
- B) Standard Deviation
- C) Mode

D) Mean

Correct Answer: Not marked

Explanation: Not available.

70. Which method is commonly used for linear dimensionality reduction?

A) DBSCAN

B) K-means clustering

C) Principal Component Analysis (PCA)

D) Hierarchical clustering

A) Decomposition

B) Linear Discriminant Analysis (LDA)

C) Principal Component Analysis (PCA)

D) K-means clustering

Correct Answer: Not marked

Explanation: Not available.

71. Which technique is commonly used for anomaly detection in time series data?

Correct Answer: Not marked

Explanation: Not available.

72. Which plot is commonly used to visualize the correlation between variables?

A) Line Plot

B) Pie Chart

C) Heatmap

D) Bar Plot

Correct Answer: Not marked

Explanation: Not available.

73. Which plot is often used in time series analysis?

A) Bar Chart

B) Line Plot

C) Scatter Plot

D) Pie Chart

Correct Answer: Not marked

Explanation: Not available.

74. What is the purpose of a box plot in EDA?

- A) To visualize the distribution of categorical data
- B) To visualize the relationship between two variables
- C) To visualize time series data
- D) To visualize summary statistics and identify outliers

Correct Answer: Not marked

Explanation: Not available.

75. What does a histogram represent?

- A) Relationship between two variables
- B) Summary statistics of a dataset
- C) Unique values in a dataset
- D) Frequency distribution of a continuous variable

Correct Answer: Not marked

Explanation: Not available.

76. Which method is used to identify the unique values in a column of a DataFrame?

- A) `uniques()`
- B) `different()`
- C) `unique()`
- D) `distinct()`

A) To fill missing values

Correct Answer: Not marked

Explanation: Not available.

77. What is the purpose of binning data in EDA?

- B) To upload data to a database
- C) To group continuous data into intervals
- D) To create visualizations

Correct Answer: Not marked

Explanation: Not available.

78. What is the purpose of data cleaning in EDA?

A) To reduce the size of the dataset

Correct Answer: Not marked

Explanation: Not available.

79. What is the main objective of clustering in EDA?

A) To upload data to a database

B) To group similar data points together based on certain features

C) To create visualizations

D) To fill missing values

Correct Answer: Not marked

Explanation: Not available.

80. How can you handle missing values in time series data?

A) Mean imputation

Correct Answer: Not marked

Explanation: Not available.

81. What is the primary purpose of data transformation in EDA?

A) To upload data to a database

B) To achieve better modeling or data analysis results

C) To reduce the size of the dataset

D) To create visualizations

Correct Answer: Not marked

Explanation: Not available.

82. Which method is commonly used for seasonal decomposition of time series?

A) STL (Seasonal and Trend decomposition using Loess)

B) Linear Discriminant Analysis (LDA)

C) Principal Component Analysis (PCA)

D) t-SNE

A) To reduce the size of the dataset

Correct Answer: Not marked

Explanation: Not available.

83. What is the primary goal of feature selection in EDA?

- B) To upload data to a database
- C) To create visualizations
- D) To select the most relevant features for modeling

Correct Answer: Not marked

Explanation: Not available.

84. What is the goal of anomaly detection in EDA?

- A) To create visualizations
- B) To upload data to a database
- C) To identify unusual data points that deviate from the norm
- D) To fill missing values

Correct Answer: A) To create visualizations

Explanation: Concept related to option A.

85. What does a scatter plot signify?

- A) The distribution of a single variable
- B) The summary statistics of a dataset
- C) The relationship between two numerical variables
- D) The count of unique values in a dataset

Correct Answer: Not marked

Explanation: Not available.

86. Which method is commonly used for feature extraction in EDA?

- A) Principal Component Analysis (PCA)
- B) Data scaling
- C) Mean imputation
- D) Data visualization

Correct Answer: Not marked

Explanation: Not available.

87. What is the IQR (Interquartile Range) used for in EDA?

- A) Measuring statistical dispersion
- B) Calculating the mean

C) Identifying the mode

D) Calculating correlation

Correct Answer: Not marked

Explanation: Not available.

88. Why might you use a 3D plot in EDA?

A) To fill missing values

B) To visualize relationships among three variables

C) To upload data to a database

D) To create databases

Correct Answer: Not marked

Explanation: Not available.

89. What is the primary goal of dimensionality reduction in EDA?

A) To reduce the complexity and size of the dataset while retaining as much information as possible

B) To fill missing values

C) To upload data to a database

D) To create visualizations

Correct Answer: Not marked

Explanation: Not available.

90. How is the mean of a dataset calculated?

Correct Answer: Not marked

Explanation: Not available.